

Compound Interest

LET – Mathematics and Statistics

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

A is the final amount

P is the starting amount

r is the annual interest rate (as a decimal)

n is the number of times the interest is compounded each year

t is the number of years

For continuous compounding:

$$A = Pe^{rt}$$

1. A saver invests £5000 in a bank account that offers a rate of interest of 4%. Calculate the amount of savings after 5 years (to the nearest penny) if:
 - (a) Interest is not compounded (simple interest) and $5 \times 4\%$ is paid
 - (b) Interest is compounded yearly
 - (c) Interest is compounded monthly
 - (d) Interest is compounded continuously
2. The value of an item is falling by 12% per year. Its price today is £250. Find the price after 8 years (to the nearest £) if:
 - (a) The depreciation is compounded quarterly
 - (b) The depreciation is compounded continuously

3. Which of the options below will result in the greatest amount of return for a saver after 10 years? [You may want to pick an amount to save, for example £1000]
- I 8% compounded yearly
 - II 7.9% compounded monthly
 - III 7.8% compounded continuously
4. A saver has the option to invest a sum of money at 12% compounded each year, for 20 years. What is the lowest continuously compounded interest rate, to 1 decimal place, that will match this option?